

CycleGAN Complete Project Execution Commands

Run these commands from the GPU cluster environment. Each command is shown separately, with suggested break points after long-running stages.

1. Go to project folder

```
cd "D:/DLP Project"
```

2. Activate environment

```
source cyclegan_env/bin/activate
```

3. Verify PyTorch and CUDA

```
python -c "import torch; print(torch.__version__); print(torch.cuda.is_available())"
```

Break point: Break here if CUDA prints False.

4. Train original baseline CycleGAN

```
python train_baseline.py
```

Break point: Break here after baseline training finishes.

5. Train modified CycleGAN with relu1_2 perceptual layer

```
python train_modified.py --vgg_layer relu1_2
```

Break point: Break here after relu1_2 training finishes.

6. Train modified CycleGAN with relu2_2 perceptual layer

```
python train_modified.py --vgg_layer relu2_2
```

Break point: Break here after relu2_2 training finishes.

7. Train modified CycleGAN with relu3_2 perceptual layer

```
python train_modified.py --vgg_layer relu3_2
```

Break point: Break here after relu3_2 training finishes.

8. Train modified CycleGAN with relu4_2 perceptual layer

```
python train_modified.py --vgg_layer relu4_2
```

Break point: Break here after relu4_2 training finishes.

9. Evaluate baseline vs chosen modified model

```
python evaluate.py --baseline_checkpoint checkpoints/baseline/baseline_epoch_200.pth  
--modified_checkpoint checkpoints/modified/modified_relu3_2_epoch_200.pth
```

Break point: Break here after evaluation finishes.

10. Run full ablation study

```
python ablation_study.py --baseline_checkpoint checkpoints/baseline/baseline_epoch_200.pth
```

Break point: Break here after ablation finishes.

11. Generate report figures

```
python visualize_results.py --baseline_checkpoint checkpoints/baseline/baseline_epoch_200.pth  
--modified_checkpoint checkpoints/modified/modified_relu3_2_epoch_200.pth
```

Break point: Final report figures are saved in results/figures/.

Resume Commands

Use these only if a training run stops early. Replace epoch_100 with the latest checkpoint epoch available in the folder.

Resume baseline

```
python train_baseline.py --resume checkpoints/baseline/baseline_epoch_100.pth
```

Resume relu1_2 modified

```
python train_modified.py --vgg_layer relu1_2 --resume  
checkpoints/modified/modified_relu1_2_epoch_100.pth
```

Resume relu2_2 modified

```
python train_modified.py --vgg_layer relu2_2 --resume  
checkpoints/modified/modified_relu2_2_epoch_100.pth
```

Resume relu3_2 modified

```
python train_modified.py --vgg_layer relu3_2 --resume  
checkpoints/modified/modified_relu3_2_epoch_100.pth
```

Resume relu4_2 modified

```
python train_modified.py --vgg_layer relu4_2 --resume  
checkpoints/modified/modified_relu4_2_epoch_100.pth
```

Expected Outputs

Stage	Output folder
Baseline samples	results/baseline/
Baseline checkpoints	checkpoints/baseline/
Modified samples	results/modified/
Modified checkpoints	checkpoints/modified/
Evaluation outputs	results/evaluation/
Ablation outputs	results/ablation/
Report figures	results/figures/